KCG COLLEGE OF TECHNOLOGY
AFFILIATED TO ANNA UNIVERSITY | AUTONOMOUS

Unit I

	COMPUTER SCIENCE AND ENGINEERING(CYBER SECURITY)		
	23CB411 COMPUTER NETWORKING		
	First		
QNo	**Questions**	**COs**	**Bloom's Level**
1.	What are the five network components ?	CO1	K1
2.	What is physical topology?	CO1	K1
3.	What is mesh topology?	CO1	K1
4.	Differentiate between bus and ring topology.	CO1	K1
5.	What is LAN?	CO1	K1
6.	Define point-point LAN.	CO1	K1
7.	What are the advantages of using wireless networks?	CO1	K1
8.	Elaborate BYOD.	CO1	K1
9.	What is a virus ?	CO1	K1
10.	What are Trojan horses?	CO1	K1
11.	Name the parts of Operating Systems.	CO1	K1
12.	What is GUI?	CO1	K1
13.	What are Banner Messages?	CO1	K1
14.	What is the difference between ports and addresses	CO1	K1
15.	List the steps for configuring ip address.	CO1	K1
16.	What is protocol layering.?	CO1	K1
17.	What is a switch?	CO1	K1
18.	Define data encapsulation .	CO1	K1
19.	What is data access?	CO1	K1
20.	What is transport layer?	CO1	K1
	Part – B		
1.	Explain about network representation and topologies.	CO1	K2
2.	Write about Internet connections with a neat flowchart.	CO1	K2
3.	Explain about security threats and solutions.	CO1	K2
4.	Explain about basic switch and end device configurations.	CO1	K2
5.	Explain about command structure in detail.	CO1	K2
6.	Write in detail about ports and addresses.	CO1	K2
7.	Explain about protocols and models	CO1	K2
8.	Describe about standards and methods in end device configurations.	CO1	K3
9.	Draw a diagram and analyse about reference models in detail.	CO1	K3

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UNIT II

QNo	Questions	COs	Bloom's Level
1	What is Ethernet?	CO2	K1
2	What is physical layer?	CO2	K1
3	Define the purpose of physical layer.	CO2	K1
4	What are the characteristics of copper cabling.	CO2	K1
5	What is coaxial cable.?	CO2	K1
6	Define UTP cabling.	CO2	K1
7	Elaborate FTTH.	CO2	K1
8	Write any 2 properties of wireless media.	CO2	K1
9	What is a number system?	CO2	K1
10	What are the types of data link layer?	CO2	K1
11	What is simplex mode?	CO2	K1
12	What is duplex mode?	CO2	K1
13	What is half-duplex mode?	CO2	K1
14	What is Ethernet encapsulation ?	CO2	K1
15	What are switch speeds ?	CO2	K1
16.	How the datalink layer prepares network data for the physical network?	CO2	K2
17.	On which two criteria the media access control method used depends on?	CO2	K2
18.	Which organization defines standards for the network access layer	CO2	K1
19.	How a wireless device sends an 802.11 frame?	CO2	K1
20.	What kind of Network would use point-to-point , hub and spoke or mesh topologies. Justify your answer.	CO2	K1
Part – B			
1	Explain about the different types of cabling in detail.	CO2	K2
2	Explain about fibre optic connectors in detail .	CO2	K3
3	Explain the properties and types of wireless media.	CO2	K2
4	Explain the steps to convert binary to hexadecimal and vice versa in detail with an example.	CO2	K2
5	Give an example for different types of topologies in detail .	CO2	K2

UNIT III

QNo	Questions	COs	Bloom's Level
1	What is a network layer?	CO3	K1
2	What is IPV4 packet?	CO3	K1
3	Give an example for IPV6 packet.	CO3	K1
4	What is host routing?	CO3	K1
5	Define static routing	CO3	K1

6	Define dynamic routing.	CO3	K1
7	What is path-vector routing.	CO3	K1
8	What is the difference between MAC and IP?	CO3	K1
9	What is ARP?	CO3	K1
10	What is a router?	CO3	K1
11	Which OSI layer sends segments to be encapsulated in an IPV4 or IPV6 packet?	CO3	K1
12.	How does IPV4 packet header indicate the message sent to be true?	CO3	K2
13	Does the time to live field used to determine the priority packet?Why?	CO3	K2
14.	Which field is used to detect corruption in the IPV4 header?	CO3	K1
15.	Which field includes common values such as ICMP (1), TCP (6), and UDP (17)?	CO3	K1
16	How many octets does the IPV6 header contain?	CO3	K1
17.	What are the features involved in default gateway router.?	CO3	K1
18.	Which two commands could be entered on a Windows host to view its IPV4 and IPV6 routing table?	CO3	K2
19	Which type of route is also known as gateway of host?	CO3	K1
20	List any two characteristics of static routes.	CO3	K1
Part – B			
1	Explain about the characteristics of IP in detail .	CO3	K2
2	Explain about the process involved in IPV4 and IPV6 in detail.	CO3	K3
3	Explain the different types of routing in detail .	CO3	K2
4	Write about distance vector routing in detail .	CO3	K2
5.	Describe about Address Resolution protocol .	CO3	K3
6.	Analyse the steps involved in basic router configuration	CO3	K2

UNIT IV

QNo	Questions	COs	Bloom's Level
1	Define the use of subnet masking.	CO4	K1
2	What is the length of subnet mask? Give an example	CO4	K1
3	What is a broadcast address?	CO4	K1
4	What is unicast address?	CO4	K1
5	Write about DMZ	CO4	K1
6	What is the need for IPV6?	CO4	K1
7	What is ICMP?	CO4	K1
8	What is the use of Ping command?	CO4	K1
9	Define RTT.	CO4	K1
10	What is Router advertisement message?	CO4	K1
11	Host-A has the IPv4 address and subnet mask 10.5.4.100	CO4	K2

	255.255.255.0. What is the network address of Host-A?		
12	Host-A has the IPv4 address and subnet mask 172.16.4.100 255.255.0.0. What is the network address of Host-A?	CO4	K2
13	Host-A has the IPv4 address and subnet mask 192.168.1.50 255.255.255.0. Which of the following IPv4 addresses would be on the same network as Host-A?	CO4	K2
14	List any two statements about private IPV4 addresses.	CO4	K1
15	Which organization or group of organizations receives IP addresses from IANA and is responsible for allocating these addresses to ISPs and some organizations?	CO4	K1
16	Which devices will not forward an IPv4 broadcast packet by default?	CO4	K1
17	What kind of subnetting masking is necessary for packet switching?	CO4	K1
18	Define VLSM.	CO4	K1
19.	What are the features of traditional subnetting?	CO4	K1
20.	List down the disadvantages of using network segmentation	CO4	K1
Part – B			
1	Explain about IPV4 Address structure in detail .	CO4	K2
2	Differentiate public and private ipv4 addresses with related examples.	CO4	K2
3	Explain about network segmentation in detail .	CO4	K2
4	Explain the steps involved in subnetting an octet boundary.	CO4	K2
5	Explain the steps to create subnets with slash 16 prefix.	CO4	K2
6	Explain about VLSM in detail .	CO4	K2
7	Explain the types of IPV6 unicast addresses.	CO4	K2
8	Analyse the structure of IPV6 GUA .	CO4	K3
9	Analyse about trace route tests connectivity.	CO4	K3

Unit V

QNo	Questions	COs	Bloom's Level
1	What is the role of transport layer?	CO5	K2

2	What is TCP?	CO5	K1
3	What is UDP?	CO5	K1
4	Write about socket pairs.	CO5	K2
5	List the 4 steps involved in session termination .	CO5	K1
6	What is a peer-peer model ?	CO5	K1
7	What is the use of GET command?	CO5	K1
8	What is IMAP?	CO5	K1
9	Elaborate DNS with an example.	CO5	K1
10	What is FTP?	CO5	K1
11	Define web and email protocols	CO6	K1
12	What is SMB?	CO6	K1
13	What is a hardware threat?	CO6	K1
14	Define maintenance threats.	CO6	K1
15	Compare the difference between worm and virus.	CO6	K2
16	List the types of firewalls.	CO6	K1
17	Give an example for creating a strong password.	CO6	K1
18	What is a webserver and a client?	CO6	K1
19	What are default gateway issues?	CO6	K1
20	What is the use of nslookup command?	CO6	K1
	Part – B		
1	Explain about different types of network attacks in detail .	CO5	K2
2	Explain about network attack mitigations.	CO5	K2
3	Explain about firewalls in detail .	CO5	K2
4	An organization plans to implement VLANs to improve network efficiency. Describe the steps involved in setting up VLANs and ensuring proper communication between them.	CO5	K4
5	A company has two branch offices connected via a WAN link. Users report slow access to resources. Identify and troubleshoot the possible causes.	CO6	K2
6	Explain the steps to verify connectivity with ping command.	CO6	K2
7.	Analyze the host and Ios commands in detail.	CO6	K3
8.	Analyse about IP Addressing Issues on End Devices.	CO6	K3

Course coordinator **HOD** **Chief Auditor**